

BIM

Execution Plan

[PROJECT NAME] — [SCOPE / WORKS PACKAGE]

([Brief scope description])

[PROJECT-NUMBER-ORIGINATOR-ZONE-LEVEL-FORM-DISCIPLINE-DOCNUMBER]

Rev.00

© *Copyright* [YEAR CLIENT NAME].

Date ([Month Year])

Prepared by: [Main Contractor Name]

Main Contractor under [General Contractor Name]

Document Code	[PROJECT-NUMBER-ORIGINATOR-ZONE-LEVEL-FORM-DISCIPLINE-DOCNUMBER]
Revision	00
Status	Issued for Approval
Date	[Month Year]
Prepared by	[Main Contractor Name]
Reviewed by	[GC BIM Manager]
Approved by	[Supervision Consultant BIM Manager / Client]

Confidentiality Statement

This document is a confidential document prepared by **[Main Contractor Name]** on behalf of **[General Contractor]** for the **[Project Name – Works Package]**. The information contained in this document is the property of and exclusively for the use of **[Client / Employer]** and authorized stakeholders. It shall not be shared with other parties without prior written consent from **[Client / Employer]**.

Obligation of Confidentiality

All recipients of this document are obliged to:

- **Maintain confidentiality:** Keep the information contained in this document confidential and disclose it only to authorized personnel involved in the project.
- **Prevent unauthorized access:** Take reasonable measures to prevent unauthorized access, copying or distribution of this document.
- **Use of Information:** Use the information provided in this document only for the purposes of this project and not for any other purpose without the explicit written consent of the Client.

Duration of Confidentiality

The obligation of confidentiality shall remain in effect for two (2) years after the completion of the project unless otherwise agreed in writing by the Client.

Breach of Confidentiality

Any breach of this confidentiality agreement may result in legal action and will be addressed in accordance with the contractual arrangements in place between the Appointed Party and the Client.

Acknowledgment

By receiving and using this document, all parties acknowledge their understanding and agreement to adhere to the terms of this confidentiality statement.

Document Management

Document History

Document Code	Revision	Prepared by	Reviewed by	Approved by
[Document Code]	00	[Main Contractor Name]	[GC BIM Manager]	

Version History

Document Code	Rev	Purpose	Rev. Date	Document Title	Document Name
[Document Code]	00	Issued for Approval	[DD/MM/YYYY]	[BIM Execution Plan – Works Package]	[Project BEP Short Name]

Note: This document is a controlled document. It is mandated to be subjected to a formal review, approval, and distribution process, which must be carried out under strictly controlled conditions.

TABLE OF CONTENTS

Confidentiality Statement	2
Obligation of Confidentiality	2
Duration of Confidentiality	2
Breach of Confidentiality	2
Acknowledgment	2
Document Management	3
Document History	3
Version History	3
TABLE OF CONTENTS	4
1 Terms and Definitions	7
1.1 General Terms	7
1.2 Acceptance Criteria	7
1.3 Terms Related to Assets and Projects	7
1.4 Terms Related to Information Management	8
2 Acronyms and Abbreviations	11
3 BIM Execution Plan Overview	13
3.1 Introduction	13
3.2 Purpose of the Document	13
3.3 Ownership and Use of Project Information	13
3.4 Deviations from Standards and EIR	13
4 Project Information	14
4.1 General Project Information	14
4.2 Project Brief	14
4.3 Project Scope	14
4.4 Project Location	15
5 Information Management	16
5.1 Project Goals and Objectives	16
Goals	16
Objectives	16
KPIs	17
5.2 Information Management	18
Information Management Parties and Teams	18
Parties and Teams	18
BIM Team	18
Mitigation Strategy	19
6 Information Requirements	20
6.1 Information Uses	20
6.2 Information Deliverables and Delivery Milestones	21
Information Delivery Milestones	21

Information Deliverables	21
6.3 Information Collaboration.....	22
Common Data Environment (CDE)	22
CDE Folder Structure.....	22
6.4 Federation Strategy and Information Container Breakdown Structure	23
6.5 Project Information Model.....	23
Level of Information Need (LOIN)	23
Level of Detail (LOD)	23
Level of Information (LOI)	23
6.6 Master Plan Requirements	24
6.7 4D and 5D Construction Sequencing.....	24
4D and 5D Technologies.....	24
6.8 Technical Requirements	24
Information Container Exchange Formats	24
Information Container Size.....	25
Information Container Coordinates	25
Information Container Units	25
6.9 Spatial Coordination Requirements	25
Spatial Coordination Strategy	25
Clash Detection Criteria	25
6.10 Information Delivery.....	26
BIM Progress Meetings.....	26
6.11 Information Quality	26
QA/QC Checklist.....	26
6.12 Information Verification and Validation.....	26
6.13 Asset Information Model	26
6.14 Project Close-Out	27
6.15 Information Security Requirements.....	27
7 Information Standards.....	28
7.1 Project Standards	28
7.2 Naming Convention	28
Information Container Naming Convention	28
Originator Code	29
Functional Breakdown by Discipline.....	29
Form / Document Type	30
Disciplines Codes List.....	30
Family Naming System	32
Materials Naming System	32
Work-sets Naming System.....	32
Information Container Metadata – Status.....	33
7.3 Classification Standards	33

7.4 Cost Classification	33
7.5 COBie Data	33
7.6 Modeling Standards.....	34
7.7 Data Authoring	34
Software	34
Hardware	34
8 Deliverables Summary	36
8.1 Resources to be Prepared.....	36
8.2 Stakeholders Responsibilities	36
9 Appendices	38
9.1 List of Appendices	38

1 Terms and Definitions

For the purposes of this document, the terms and definitions given in ISO 19650-1, ISO 19650-2 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

1.1 General Terms

Responsibility matrix

The below table outlines and defines the roles and responsibilities of each project team member regarding the use and implementation of BIM throughout the project lifecycle.

SN	Organization	Role
1	[Client / Employer]	Client / Employer
2	[PMC]	Project Management Consultant
3	[Design Consultant]	Designer
4	[Supervision Consultant]	Supervision Consultant
5	[General Contractor]	General Contractor
6	[Main Contractor Name]	[Main Contractor – [Scope]]

Table 1 – Responsibility Matrix for the BIM and Engineering Teams

1.2 Acceptance Criteria

[Main Contractor Name] is the appointed Main Contractor under [General Contractor] for the [Works Scope] within the [Project Name] project.

[Main Contractor Name] accepts and complies with all requirements provided by [Client] and [General Contractor] as per contractual obligations and the Exchange Information Requirements (EIR).

1.3 Terms Related to Assets and Projects

Actor

Person, organization or organizational unit involved in a construction process.

Appointment

Agreed instruction for the provision of information concerning works, goods or services.

Appointed party

[Definition of Appointed party in context of this project]

Appointing party

[Definition of Appointing party in context of this project]

Client

[Definition of Client in context of this project]

Delivery team

[Definition of Delivery team in context of this project]

Task team

[Definition of Task team in context of this project]

Asset

Item, thing or entity that has potential or actual value to an organization.

Project information

Information produced for, or utilized in, a particular project.

Life cycle

Life of the asset from the definition of its requirements to the termination of its use, covering its conception, development, operation, maintenance support and disposal.

Delivery phase

Part of the life cycle during which an asset is designed, constructed and commissioned.

Operational phase

Part of the life cycle during which an asset is used, operated and maintained.

Trigger event

Planned or unplanned event that changes an asset or its status during its life cycle, which results in information exchange.

Key decision point

Point in time during the life cycle when a decision crucial to the direction or viability of the asset is made.

Project team

Appointing party and all delivery teams.

Plan of work

Document that details principal stages in the design, construction work and maintenance of a project and identifies the main tasks and people.

Stage

Distinct period in a project used as a management tool.

1.4 Terms Related to Information Management

Information

Reinterpretable representation of data in a formalized manner suitable for communication, interpretation or processing.

Information requirement

Specification for what, when, how and for whom information is to be produced.

Organizational information requirements (OIR)

Information requirements in relation to organizational objectives.

Asset information requirements (AIR)

Information requirements in relation to the operation of an asset.

Project information requirements (PIR)

Information requirements in relation to the delivery of an asset.

Exchange information requirements (EIR)

Information requirements in relation to an appointment.

Information exchange

Act of satisfying an information requirement or part thereof.

Information model

Set of structured and unstructured information containers.

Asset information model (AIM)

Information model relating to the operational phase.

Project information model (PIM)

Information model relating to the delivery phase.

Federation

Creation of a composite information model from separate information containers.

Information container

Named persistent set of information retrievable from within a file, system or application storage hierarchy.

Status code

Meta-data describing the suitability of the content of an information container.

Building information modelling (BIM)

Use of a shared digital representation of a built asset to facilitate design, construction and operation processes to form a reliable basis for decisions.

Common data environment (CDE)

Agreed source of information for any given project or asset, for collecting, managing and disseminating each information container through a managed process.

Level of information need

Framework which defines the extent and granularity of information.

Capability

Measure of ability to perform and function.

Capacity

Resources available to perform and function.

BIM execution plan (BEP)

[Definition of BIM execution plan (BEP) in context of this project]

Information delivery milestone

Scheduled event for a predefined information exchange.

Master information delivery plan (MIDP)

Plan incorporating all relevant task information delivery plans.

Task information delivery plan (TIDP)

Schedule of information containers and delivery dates, for a specific task team.

Asset management

Coordinated activity of an organization to realize value from assets.

Facility management

Organizational function which integrates people, place and process within the built environment with the purpose of improving the quality of life of people and the productivity of the core business.

Information provider

Actor who provides information in an information container.

Information receiver

Actor who receives information in an information container.

Information reviewer

Actor who reviews information and its information container.

Metadata

Data about data.

Need-to-know

Legitimate requirement of a prospective recipient of information to know, to access, or to possess sensitive information.

Risk appetite

Amount and type of risk that an organization is willing to pursue or retain.

Safety

State of relative freedom from threat or harm caused by random, unintentional acts or events.

Security

State of relative freedom from threat or harm caused by deliberate, unwanted, hostile or malicious acts.

Security breach

Infraction or violation of security.

Security incident

Suspicious act or circumstance threatening security.

Security-minded

Understanding and routinely applying appropriate and proportionate security measures in any business situation so as to deter and/or disrupt hostile, malicious, fraudulent and criminal behaviours or activities.

Sensitive information

Information, the loss, misuse or modification of which, or unauthorized access to, can adversely affect the privacy, security or safety of an individual or individuals; compromise intellectual property or trade secrets of an organization; cause commercial or economic harm to an organization or country; and/or jeopardize the security, internal and foreign affairs of a nation.

Residual risk

Risk that remains after controls have been implemented.

Threat

Potential cause of an incident which may result in harm.

Top management

Person or group of people who directs and controls an organization at the highest level.

Vulnerability

Weakness that can be exploited to cause harm.

2 Acronyms and Abbreviations

Abbreviation	Full Term	Abbreviation	Full Term
AIM	Asset Information Model	IT	Information Technology
AIR	Asset Information Requirements	ITT	Invitation to Tender
AMS	Asset Management System	KPI	Key Performance Indicators
BCF	BIM Collaboration Format	LDC	Lead Design Consultant
BEP	BIM Execution Plan	LOD	Level of Detail
BIM	Building Information Modelling	LOI	Level of Information
BOQ	Bill of Quantity	LOIN	Level of Information Need
BSI	British Standards Institute	MIDP	Master Information Delivery Plan
CAD	Computer Aided Design	MPDT	Model Production Delivery Table
CAFM	Computer Aided Facility Management	NRM	New Rules of Measurement
CC	Cost Consultant	O&M	Operations and Maintenance
CDE	Common Data Environment	OIR	Organizational Information Requirements
COBie	Construction-Operations Building Information Exchange	PIM	Project Information Model
CSI	Construction Specification Institute	PIR	Project Information Requirements
DC	Design Consultant	POW	Plan of Work
EDMS	Electronic Document Management System	QA	Quality Assurance
EIR	Exchange Information Requirements	QC	Quality Control
FM	Facilities Management	RACI	Responsible, Accountable, Consulted, Informed
GC	General Contractor	RIBA	Royal Institute of British Architects
GIA	Gross Internal Area	SAMP	Strategic Asset Management Plan
GIS	Geographic Information System	SSC	Site Supervision Consultant
IFC	Industry Foundation Classes	TIDP	Task Information Delivery Plan
ISO	International Standards Organization	WIP	Work in Progress

Table 2 – Acronyms and Abbreviations

[Add project-specific acronyms as needed]

3 BIM Execution Plan Overview

3.1 Introduction

Building Information Modelling (BIM) is a process focused on the development, use and sharing of information-rich digital models in order to improve the design, construction and operations of a project or portfolio of facilities.

This BIM Execution Plan (BEP) has been prepared by **[Main Contractor Name]** as the appointed Main Contractor under **[General Contractor]** for the **[Works Package]** component of the **[Project Name]** project at **[Location, City]**. The **[Works Package]** scope covers **[brief scope description, e.g. all external hardscape elements outside plot boundaries]**. This BEP is dedicated to the Architecture, Structure, and MEP (Mechanical, Electrical & Plumbing) disciplines, and defines the discipline-wise model breakdown strategy applied across all delivery stages.

The BEP provides a framework so that the project team members and any others involved in the project understand the information delivery requirements, the organization and members of the project team, and the structure of the project files. The BEP is a live document that should be updated in every project phase — Construction stage and As-Built stage — to serve as a central reference for all parties and is to be shared with all project team members.

3.2 Purpose of the Document

This BIM Execution Plan (BEP) is a foundational document that outlines how Building Information Modelling (BIM) will be implemented throughout the lifecycle of the **[Works Package]** at **[Project Name]**. Its primary purpose is to define roles, responsibilities, standards, workflows, deliverables, and information exchange protocols to ensure consistent, coordinated, and efficient use of BIM among all stakeholders.

Developed in alignment with the Exchange Information Requirements (EIR) issued by **[Client]** and the ISO 19650 standard, this BEP ensures that the **[Main Contractor Name]** team works collaboratively with **[GC, Supervision Consultant]**, and all relevant stakeholders using a common framework for data management and model development.

3.3 Ownership and Use of Project Information

[Client] retains full ownership of all information produced for the project, including but not limited to the encompassed data and all related copyrights. This ownership extends to the unrestricted use of such data by **[Client]** following completion of any project component or phase.

[Main Contractor Name], under the direction of **[General Contractor]**, is committed to managing project information in accordance with contractual obligations, intellectual property laws, and ISO 19650 information management standards. All project-related data — including models, documents, and metadata — are developed and shared within a secure Common Data Environment (CDE), specifically **[CDE Platform (e.g. Autodesk Construction Cloud / Aconex)]**, to ensure traceability, version control, and authorized access.

3.4 Deviations from Standards and EIR

[Main Contractor Name] will adhere to all established standards and Employer's Information Requirements (EIR) without any deviations. Any proposed deviation, should it become necessary due to unforeseen technical or contractual reasons, will be formally submitted to **[General Contractor]** and subsequently to **[Supervision Consultant and Client]** for review and approval prior to implementation.

No deviations are currently anticipated or planned for this scope.

4 Project Information

4.1 General Project Information

General Project Information	
Project Name	[Project Name]
Project Address	[Full Project Address]
Project Number	[P####-K#####]
BEP Document Code	[Document Code]
Project Type	[e.g. Residential / Commercial / Infrastructure]
Scope Covered by This BEP	[Works Package Description]
Project CDE	[e.g. ACC (Autodesk Construction Cloud), Aconex]
Project Stage	Construction
Project Duration	[From DD-MM-YYYY to DD-MM-YYYY]
Construction Commencement Date	[DD-MM-YYYY]
As-Built Completion Date	[DD-MM-YYYY]
Main Contractor	[Main Contractor Name]
Lead Appointed Party	[General Contractor Name]
Supervision Consultant	[Supervision Consultant Name]
Client / Employer	[Client Name]

Table 3 – General Project Information

4.2 Project Brief

[Describe the project location, development context, and master plan overview. Include the project's purpose, scale, and key components (e.g. number of units, land uses, precincts). Minimum 2–3 paragraphs.]

4.3 Project Scope

[Describe the overall project scope (e.g. built-up area, number of floors/blocks). This BEP covers the Architecture, Structure, and MEP disciplines for the works package described below.]

The scope covered by this BEP includes:

- [Architecture – scope description]
- [Structure – scope description]
- [MEP (Mechanical, Electrical & Plumbing) – scope description]
- [BIM deliverables: Models to LOD 400 during construction, LOD 500 for As-Built, broken down by discipline]

4.4 Project Location

[Full address and context. Describe how the project site is accessible and how the works scope relates to the wider masterplan or development.]

[Insert Figure – Project Location & Masterplan]

5 Information Management

5.1 Project Goals and Objectives

Goals

In alignment with **[Client's]** vision and commitment to leading high-quality development through Building Information Modelling (BIM), the following goals have been set to guide the implementation of BIM for the **[Works Package]**.

#	Goal	Description
01	Smart City Development	Foster smart city concepts through the scope lifecycle, using BIM and Digital Twins.
02	Collaborative Project Delivery	Promote continuous collaboration and coordination across all teams.
03	Seamless Planning and Construction	Employ advanced BIM technologies to streamline the planning and construction process.
04	Cost Efficiency	Achieve measurable cost savings by implementing efficient information management processes.
05	Standardized BIM Processes	Embed BIM information management processes as standard practice across all deliverables.
06	Data Driven Decision Making	Ensure access to accurate, timely project information to facilitate informed decision-making.
07	Optimized Information Utilization	Leverage accurate information for various applications, benefiting all stakeholders.
08	Transparent Deliverable Sign-off	Use information to drive a transparent and accountable sign-off process for all deliverables.
09	Comprehensive Information Handover	Ensure that at completion of works, usable and comprehensive information is handed over.
10	Continuous Learning and Improvement	Learn lessons collectively on the use of BIM and record insights for future reference.

Table 4 – Project Goals

Objectives

#	Goal	Objective
01	Smart City Development	Develop and implement a strategic plan to integrate BIM and Digital Twins in all scope elements, with the objective of implementing smart solutions aligned with the Client's vision.
02	Collaborative Project Delivery	Establish a comprehensive collaboration protocol using BIM, aiming to increase team integration, measured by reduced conflicts and enhanced team satisfaction.

03	Seamless Planning and Construction	Train the project team in advanced BIM technologies to reduce planning and construction timelines, enhancing efficiency and reducing errors.
04	Cost Efficiency	Employ BIM and related technologies to streamline workflows, aiming to achieve a quantifiable reduction in overall project costs.
05	Standardized BIM Processes	Achieve 100% adherence to standardized BIM processes for all deliverables within the project timeline.
06	Informed Decision Making	Create a real-time project information dashboard that ensures timely and accurate information flow, leading to improved decision-making efficiency.
07	Optimized Information Utilization	Utilize BIM for various applications across the scope lifecycle, aiming to increase overall project value and stakeholder satisfaction.
08	Transparent Deliverable Sign-off	Implement a digital tracking and approval system for all deliverables, aiming to make the sign-off process 100% transparent.
09	Comprehensive Information Handover	Develop and implement a standardized information handover protocol ensuring all completed works transfer comprehensive and usable information.
10	Continuous Learning and Improvement	Establish a knowledge management system to capture and share lessons learned from BIM usage, with the aim of continuously improving project outcomes.

Table 5 – Project Objectives

KPIs

#	Goal / Objective	KPI Description	Measurement Metric
01	Smart City Development	Number of smart features integrated	Count of implemented features
02	Collaborative Delivery	Coordination issue resolution time	Percentage of issues resolved
03	Seamless Planning	Reduction in design errors	Percentage decrease in errors
04	Cost Efficiency	Documented cost savings	Cost savings percentage
05	Standardized BIM	Compliance with BIM standards	Compliance rate percentage
06	Informed Decision Making	Decisions supported by BIM data	Number of decisions
07	Optimized Info Use	Benefits from BIM applications	Qualitative and quantitative benefits
08	Transparent Sign-off	Time from submission to approval	Days or hours taken
09	Info Handover	Completeness of BIM data at handover	Completeness percentage
10	Continuous Learning	Lessons learned applied	Number of improvements

Table 6 – Project KPIs

5.2 Information Management

Information Management Parties and Teams

Project Stakeholders	
Appointing Party	[Client Name]
PMC	[PMC Name]
Lead Design Consultant	[Design Consultant Name]
Supervision Consultant	[Supervision Consultant Name]
General Contractor	[General Contractor Name]
Main Contractor	[[Main Contractor Name] – [Scope]]

Table 7 – Project Stakeholders

Parties and Teams

Organizational Role	Name	Organization	Email
[Role]	[Name]	[Client]	[email@client.com]
[Role]	[Name]	[PMC]	[email@pmc.com]
[Role]	[Name]	[Supervision Consultant]	[email@sc.com]
[Role]	[Name]	[General Contractor]	[email@gc.com]
[Main Contractor] Project Manager	[Name]	[Main Contractor Name]	[email@maincontractor.com]
[[Main Contractor] Lead Engineer / BIM Coordinator]	[Name]	[Main Contractor Name]	[email@maincontractor.com]

Table 8 – Parties and Teams Contacts

BIM Team

The [Main Contractor Name] team consists of:

- [Main Contractor] Project Manager – Overall oversight of BIM deliverables, compliance with BEP, coordination with GC BIM Manager
- [Main Contractor] Lead Engineer – Day-to-day model management, coordination across disciplines, clash detection, QA/QC
- [Main Contractor] BIM Coordinator – Model authoring, workset management, CDE uploads, naming convention compliance
- BIM Technicians / Modelers – Production of BIM elements per LOD requirements, updating models during construction

Mitigation Strategy

Code	[EIR Reference Code]
Purpose	Identify and address potential risks associated with information delivery.
Requirement	[[Main Contractor Name] shall develop a mitigation strategy in coordination with the GC.]
Deliverable	A comprehensive mitigation strategy outlining measures to address identified risks.

Table 9 – Mitigation Strategy Schedule

[Main Contractor Name] will implement a structured risk mitigation strategy in accordance with ISO 19650 for managing information and project delivery. This strategy ensures that risks related to project execution, digital delivery, and information management are proactively identified, assessed, mitigated, and monitored throughout the project lifecycle.

6 Information Requirements

6.1 Information Uses

[Review and update ☒ ☐ checkboxes below to match this project's scope]

Code	Information Use	Concept	Design Development	Construction	As-Built
IU-01	Existing Conditions Modelling	☐	☐	☐	☐
IU-02	Site Capturing Operation	☐	☐	☒	☒
IU-03	As-Built Modelling	☐	☐	☒	☒
IU-04	Master Plan Model Authoring	☐	☐	☒	☒
IU-05	Project Information Model Authoring	☐	☐	☒	☒
IU-06	Temporary Construction Systems Model Authoring	☐	☐	☒	☐
IU-07	Construction Site Logistics Model Authoring	☐	☐	☒	☐
IU-08	Asset Information Model Authoring	☐	☐	☐	☒
IU-09	Construction System Design	☐	☐	☒	☒
IU-10	Site Utilization Planning	☐	☐	☒	☐
IU-11	Design Review	☐	☐	☒	☐
IU-12	Drawings Production	☐	☐	☒	☐
IU-13	Engineering Analysis	☐	☐	☒	☐
IU-14	Structural Analysis	☐	☐	☐	☐
IU-15	Energy Analysis	☐	☐	☐	☐
IU-16	Sustainability Analysis	☐	☐	☒	☐
IU-17	Visualization	☐	☐	☒	☐
IU-18	3D Coordination	☐	☐	☒	☐
IU-19	3D Control and Planning	☐	☐	☒	☒
IU-20	3D Digital Fabrication	☐	☐	☒	☐
IU-21	4D Phase Planning	☐	☐	☒	☐
IU-22	4D Construction Simulation	☐	☐	☒	☐
IU-23	5D Cost Estimation and Modelling	☐	☐	☒	☐
IU-24	5D Cost Analysis	☐	☐	☒	☒

IU-25	Augmented Reality and Virtual Reality	☐	☐	☒	☒
IU-26	Information Validation	☐	☐	☒	☒
IU-27	Deliverables Progress Tracking	☐	☐	☒	☒
IU-28	Issue Tracking	☐	☐	☒	☒
IU-29	Request for Information	☐	☐	☒	☒
IU-30	Work Inspection	☐	☐	☒	☒
IU-31	eSnagging	☐	☐	☐	☒
IU-32	Maintenance Scheduling	☐	☐	☐	☐
IU-33	System Performance Analysis	☐	☐	☐	☐
IU-34	Space Management and Tracking	☐	☐	☒	☒
IU-35	Asset Management	☐	☐	☒	☒
IU-36	Asset Information Registry	☐	☐	☒	☒

Table 10 – Information Uses

6.2 Information Deliverables and Delivery Milestones

Information Delivery Milestones

Information Delivery Milestones	Start Date	End Date
Manufacturing and Construction (LOD 400)	[DD Month YYYY]	To be assigned per approved Baseline Program
Handover / As-Built (LOD 500)	To be assigned per approved Baseline Program	[DD-MM-YYYY]

Table 11 – Information Delivery Milestones

Information Deliverables

Information Deliverables	Manufacturing & Construction	Monthly Submission	As-Built
Post-appointment BIM Execution Plan	✓	•	✓
Contribute to Project Risk Register	✓	•	✓
Master Information Delivery Plan (MIDP)	✓	✓	✓
Model Production Delivery Table (MPDT)	✓	•	✓
Information Management Assignment Matrix (RACI)	✓	•	✓

Level of Information Need	✓	✓	✓
Clash Matrix	✓	✓	✓
QA/QC Check List	✓	✓	✓
Contribute to Project Lessons Learnt	✓	•	✓
Asset Register	✓	✓	✓
BIM Progress Report	✓	✓	✓
Coordination Report	✓	✓	✓
Project Information Model	✓	✓	✓
Federated Model	✓	✓	✓
Asset Information Model	•	•	•

Table 12 – Information Deliverables

6.3 Information Collaboration

Common Data Environment (CDE)

The CDE for the [Works Package] provides a single source of truth for all project information, hosted on [Autodesk Construction Cloud (ACC) / BIM 360 / Common Data Environment Platform Name]. The CDE will be structured to support discipline-wise information delivery (Architecture, Structure, MEP) in alignment with construction sequencing and submission strategy, and follows the four-state container workflow defined in ISO 19650-1: Work in Progress (WIP), Shared, Published, and Archive. **[Works Package]** provides a single source of information for the project. The CDE will be structured to support **[discipline-wise]** information delivery in alignment with construction sequencing and submission strategy.

- All BIM-related information shall be hosted in the CDE; no discipline model or document shall be exchanged outside this environment.
- Project team members shall strictly follow the CDE protocols, including status codes, naming conventions, and approval workflows defined in this BEP.
- Access to the CDE shall be role-based, with permissions assigned per discipline team (Architecture, Structure, MEP) and reviewed at each project stage.
- Information containers shall move between CDE states (WIP to Shared to Published to Archive) only after passing the QA/QC checks and approval workflow defined in Section 6.11.

CDE Folder Structure

SN	ISO Format ACC Folder Structure	GC / [Main Contractor] Linked Folder Structure
1	01. Work In Progress folder	[00. WIP Models – [Scope] (Discipline-wise)]
2	02. Shared – Collaboration	[100. Shared (Discipline-wise)]
3	Published	[01. Submission – [Scope] (Discipline-wise)]
4	Archive	[02. Archive (Discipline-wise)]

Figure 5 – CDE Folder Structure

6.4 Federation Strategy and Information Container Breakdown Structure

[Describe the federation strategy for this project – how discipline models are aligned, what each information container covers, and the coordination workflow.]

Information Container	Comments
[Main Discipline / Works Container]	[Description of elements included]
[Sub-Discipline Container]	[Description of elements included]
Coordination Model	Interface with other disciplines

Table 13 – Model Information Container

6.5 Project Information Model

[Main Contractor Name] will follow the size criteria for modelling elements: any element larger than 25 mm will be modelled; any element less than that will not be modelled, unless the element covers a wide range and might have a potential clash with other elements.

Level of Information Need (LOIN)

Information Delivery Milestone	Level of Information Need (LOIN)
IFC / Issued for Construction	300/350 (LOD 300/350 + LOI 300/350)
Manufacturing and Construction	400 (LOD 400 + LOI 400)
As-Built	500 (LOD 500 + LOI 500)

Table 14 – Level of Information Need by Delivery Milestones

Level of Detail (LOD)

[Describe the LOD requirements for each discipline within this works scope. State which elements are modelled by the Main Contractor and which are coordination references only.]

Level of Information (LOI)

Stage	Typical LOI Requirements
Design (Stage 3–4)	Product types, material specs, system performance, engineering ratings.
Construction (Stage 5)	Installation data, supplier/manufacture info, health & safety data, maintenance access requirements.
Handover (Stage 6)	As-built product data, serial numbers, warranty details, O&M manuals, commissioning data.
Operation (Stage 7)	Full asset data for CAFM systems, lifecycle cost info, inspection schedules, and replacement plans.

Table 15 – LOI Requirements During the Project Life Cycle

6.6 Master Plan Requirements

[Describe the project coordinate framework. Include the shared base point coordinates and any zone boundary reference points relevant to this project.]

Point	North	East	Notes
[P1]	[N coordinate]	[E coordinate]	[Zone / Boundary reference]
[P2]	[N coordinate]	[E coordinate]	[Zone / Boundary reference]

[Add additional boundary points as needed]

6.7 4D and 5D Construction Sequencing

4D Construction Sequencing will be utilized as a key deliverable. 3D models of all disciplines will be appended to the scheduling software.

Activity ID parameters for all elements will be updated with appropriate information from the baseline program. Predefined selection sets will be created to attach elements, then linked using the TimeLiner function to the planned start and end dates from the project construction program in CSV format exported from Primavera or similar software.

4D and 5D Technologies

Synchro and CostX will be used as tools for 4D–5D reporting. An updated baseline will be imported to Synchro to show, on a monthly basis, the Actual vs. Planned site progress. Task costs will be shown where applicable.

6.8 Technical Requirements

Information Container Exchange Formats

Exchange Format	Description
.rvt	Revit Models
.nwc	Navisworks Cache Files
.nwd	Navisworks (Federated Model)
.dwg	AutoCAD Drawings
.dwfx	AutoCAD Design Review
.ifc	Industry Foundation Classes
.pdf	BEP, Reports, Drawings
.HTML	Clash Reports
.sp	Bentley Synchro
.xlsx	COBie Data, Schedules

Table 16 – Data Exchange Formats

Information Container Size

Individual information containers shall not exceed 350 MB in size, with a working target of 300 MB as preferred practice. Models will be divided in the following order to maintain consistency: Discipline > Building/Zone > Phase.

Information Container Coordinates

Parameter	Value
North/South (N/S)	[N/S coordinate value]
East/West (E/W)	[E/W coordinate value]
Elevation (Z)	[0 or project datum elevation]
Angle to True North	[0.00° or project-specific angle]
Shared Coordinates Used	Yes

Figure 9 – Model Project Base Point

Information Container Units

Measure	Units
Length	mm (millimetres) – Revit / m (metres) – Civil 3D
Area	m ² (metres squared)
Volume	m ³ (metres cubed)
Angle	° (degrees)

Table 17 – Information Container Units

6.9 Spatial Coordination Requirements

Spatial Coordination Strategy

[Main Contractor Name] Discipline Leads will be responsible for performing clash detection between all discipline models during the construction phase. Regular BIM coordination meetings will be held fortnightly. Navisworks Manage 2024 will be used for clash detection.

Clash Detection Criteria

LOD 400 Models (Construction Models)	
[Discipline vs. Discipline]	5 mm
LOD 500 Models (As-Built Models)	
All Disciplines	Acceptable clash report generated using Zero mm tolerance

Table 18 – Construction Tolerance

Stage	Tolerance (mm)
Schematic Design	70 mm
Detail Design	50 mm
Tender / IFC	25 mm
Construction	5 mm

Table 19 – Tolerance Definition / Acceptance in Each Stage

6.10 Information Delivery

BIM Progress Meetings

Regular BIM progress meetings will be scheduled (weekly, or as necessary) and will cover design reviews, coordination efforts, clash avoidance, and resolution strategies.

Meeting Participants	
Appointing Party	[Client Name]
PMC	[PMC Name]
Supervision Consultant	[Supervision Consultant Name]
General Contractor	[GC Name]
Main Contractor	[Main Contractor Name]

Table 20 – Meeting Participants

6.11 Information Quality

QA/QC Checklist

The Project Model Audit Form can be found in Appendix 02 – QA/QC Check List Template. This should be used as a guide by the project team to check model compliance with the project BIM plan and office BIM standards.

6.12 Information Verification and Validation

Level 1 – Self-Checks: Each team member is responsible for ensuring compliance with BIM Standards.

Level 2 – Peer and/or Discipline Leads: Discipline BIM leads will perform QA/QC BIM audits on a fortnightly basis.

Level 3 – Discipline Heads (Squad Checks): Each discipline BIM lead and coordinator will perform a final QA/QC BIM audit at scheduled squad checks.

Level 4 – [Main Contractor] BIM Manager: The BIM Manager will review the final Revit files for compliance to the agreed standard and the BEP before issue to **[GC]** and the supervision team.

6.13 Asset Information Model

The Asset Information Model (AIM) will adhere to relevant standards including ISO 19650 and the ISO 55000 series to ensure a comprehensive digital representation of all assets.

For As-Built works, surveys including total station measurements and LiDAR scanning (where applicable) will be used to verify and update model elements to LOD 500.

6.14 Project Close-Out

The project close-out process ensures all final deliverables are submitted, reviewed, and handed over in accordance with the Client's standards. This includes:

- Final Redline drawings
- As-Built drawings and federated models (LOD 500)
- Asset Information Model (AIM) with linked COBie data
- All documents submitted through the CDE and validated for accuracy

6.15 Information Security Requirements

To protect project data, the following security measures must be enforced:

- Secure access control within the CDE
- Role-based permissions and user authentication
- Regular backup protocols and disaster recovery plans
- Protection of sensitive and classified information in accordance with ISO standards

7 Information Standards

7.1 Project Standards

Standard / Reference	Description
Client EIR	[Client] Employer's Information Requirements – [Version / Date]
Local Authority BIM Guidelines	[e.g. Abu Dhabi Municipality BIM Guidelines – Version / Date]
GC Pre-BEP	[Document Reference – GC Pre-BEP]
CPIx Standard BEP	CPIx post contract BIM execution plan R 1.0
PAS 1192-6	Specifies requirements for the collaborative sharing of structured H&S information.
BS 1192-4:2014	Collaborative production of architectural, engineering and construction information.
ISO 19650-1	Concepts and principles for information management using BIM.
ISO 19650-2	Information management using BIM – Part 2: Delivery phase of assets.
ISO 19650-3	Information management using BIM – Part 3: Operational phase of assets.
ISO 19650-4	Information management using BIM – Part 4: Information Exchange.
ISO 19650-5	Security-minded approach to information management using BIM.
Uniclass 2015	Classification system for organizing construction information across all project stages.
ISO 27001	Information technology – security techniques – information security management systems.
RICS NRM 1	Order of cost estimating and cost planning for capital building works.
RICS NRM 2	Detailed measurement for building works.
RICS NRM 3	Order of cost estimating and cost planning for building maintenance works.
ISO 7817-1:2024	Building Information Modelling – Level of Information Need – Part 1: Concepts and Principles.
ISO 16739	Industry Foundation Classes (IFC) for data sharing in the construction and facility management industries.

Table 21 – Project Standards

7.2 Naming Convention

Information Container Naming Convention

[Update the naming convention structure below to match the project's document numbering system]

Project No.	-	Package No.	-	Originator	-	Discipline Breakdown	-	Level	-	Form	-	Discipline	-	Doc No.
-------------	---	-------------	---	------------	---	----------------------	---	-------	---	------	---	------------	---	---------

Table 22 – Models Naming Convention

Field	Code	Description
Project Number	[P####]	Fixed – indicates the project
Package Number	[K#####]	Fixed – indicates the package
Discipline Breakdown	[DS-XXX]	[Specifies the discipline model, e.g. Architecture, Structure, MEP]
Spatial Breakdown / Level	[NN]	Specifies the spatial aspect (e.g. ZZ = multiple levels, XX = not applicable)
Form / Document Type	[AA]	Describes the nature or format of the information container
Discipline Code	[AA]	Indicates the technical discipline
Document Number	[NNNNNN]	A sequential number assigned to make the ID unique

Table 22A – Naming Convention Description

Originator Code

Code	Description
[GC Code]	[General Contractor Name]
[Client Code]	[Client Name]
[DC Code]	[Design Consultant Name]
[SC Code]	[Supervision Consultant Name]
[Main Contractor Code]	[Main Contractor Name]

Table 23 – Originator Code Schedule

Functional Breakdown by Discipline

Discipline Breakdown Code	Description
DS-ALL	[Works Package – All Disciplines]
DS-ARC	[Architecture Discipline Model]
DS-STR	[Structure Discipline Model]
DS-MEP	[MEP (Mechanical, Electrical & Plumbing) Discipline Model]

Table 24 – Functional Breakdown by Discipline

Form / Document Type

Code	Description	Code	Description
M3	3D Model file	SD	Shop Drawing
M2	2D Model file	AB	As-Built
BQ	Bill of Quantities	RP	Report
RC	Clash Report	MM	Meeting Minutes
IR	Inspection Report	WI	Work Inspection Request
IT	Inspection Test Plan	PL	Plan
MS	Material Submission	RM	Monthly Report
RW	Weekly Report	QA	QA/QC Check List

Table 25 – Form / Document Type

Disciplines Codes List

AR	Architecture
ST	Structure
IN	Interior Design
FA	Facade / Envelope Engineering
AC	Acoustics
ME	Mechanical
EL	Electrical
PH	Public Health / Plumbing
FP	Fire Protection
IT	ICT / Low Current Systems
AV	Audio Visual
SC	Security Systems
BA	Building Automation (BMS)
VT	Vertical Transportation
CW	Civil Works
RW	Road Works
DR	Drainage System
SW	Stormwater
IR	Irrigation

LA	Landscape Architecture
UD	Urban Design
GE	Geotechnical
MW	Marine Works
PW	Potable Water
FW	Foul Water / Sewerage
PWR	Power Networks (HV / LV)
TC	Telecommunications
GN	Gas Networks
DC	District Cooling
SG	Signage & Wayfinding
SL	Street Lighting
LI	Lighting
SU	Survey
GIS	GIS / Geospatial
BM	BIM Management
QM	Quality Management
HS	Health & Safety
EN	Environmental
PM	Project Management
CM	Cost Management / Quantity Surveying
PG	Programme / Planning & Scheduling
FF	FF&E (Furniture, Fixtures & Equipment)
WM	Waste Management
TR	Traffic Engineering
FK	Food Service / Kitchen Design
PF	Pools & Water Features
ZZ	Multi-discipline / Whole Project
M3	3D Model
M2	2D Model / Drawing
DR	Drawing
RP	Report
PL	Plan / Schedule
QA	QA/QC Document

SH	Schedule
SP	Specification
CO	Correspondence
SD	Shop Drawing
RC	Coordination / Clash Report
RM	Progress Report

Table 26 – Disciplines Codes List

[Update discipline codes to match this project's scope]

Family Naming System

Originator	Discipline	Object Code	Subtype	Manufacturer / Generic	Descriptor
[GC Code]	[Disc.]	[OBJ]	[Subtype]	[Generic]	[Material/Desc.]

Table 27 – Family Naming System

Eg. GC Code_ Disc._ OBJ_Subtype_ Generic_ Material/Desc.

Materials Naming System

Object Type / Category	Description-01	Description-02 (Optional)
[e.g. Concrete Pavers]	[e.g. 600 x 300 mm]	[e.g. Grey]

Table 28 – Materials Naming System

Eg. Concrete Pavers_600 x 300 mm_ Grey

Work-sets Naming System

Originator	Discipline	Description
[GC Code]	[Disc. Code]	[WORKSET DESCRIPTION]

Table 29 – Work-sets Naming System

Eg. GC Code_ Disc. Code_ WORKSET DESCRIPTION

Information Container Metadata – Status

Code	Description
A	Approved
B	Approved with Comments
C	Revise and Re-submit
D	Rejected

Table 31 – Information Container Status Code

Delivery Stage	Revision Code at Submission	Revision Code During WIP
Issued for Construction	00	
Construction Phase	01	02
As-Built	AB	

Table 32 – Revision Code Structure

7.3 Classification Standards

[Main Contractor Name] confirms that all information containers will be assigned appropriate classification codes in alignment with Uniclass 2015. Key classification codes for this project scope:

- [Ss_XX_XX_XX – Description]
- [Pr_XX_XX_XX – Description]
- [Add project-specific Uniclass codes]

7.4 Cost Classification

[Main Contractor Name] will implement NRM 1 and NRM 3 to ensure accurate cost estimation and planning. All modelled elements will be structured and coded to support quantity extraction directly from the information containers in alignment with NRM standards.

7.5 COBie Data

#	COBie Activity	Client	SC	GC	[Main Contractor Code]
01	Define COBie data requirements (EIR scope)	A/R	I	I	I
02	Produce COBie data from BIM model	I	C	A	R
03	Populate non-geometric COBie fields (O&M, warranties)	I	C	A	R
04	QA/QC review of COBie data	C	A	C	R

05	Upload COBie file to CDE for review	I	C	A	R
06	Review and comment on COBie submission	A	R	C	I
07	Resolve comments and resubmit	I	C	A	R
08	Final approval of COBie data	A/R	I	C	I
09	Integrate COBie into Asset Information Model	A/R	I	C	I

Table 33 – COBie Responsibility Matrix

7.6 Modeling Standards

[Main Contractor Name] will ensure that all BIM modelling activities follow clearly defined guidelines to maintain accuracy, consistency, and efficiency throughout the construction and handover phases.

- All elements must be modelled in 3D using the agreed software, adhering to specified LOD and LOI levels as outlined in this BEP
- Model ownership responsibilities are clearly assigned per the RACI matrix
- All modelling must comply with the project federation strategy, container breakdown structure, and revision control procedures

7.7 Data Authoring

Software

BIM Activity	Software	Version
Design Authoring (Landscape / Hardscape)	Autodesk Revit	2024
Design Authoring (Civil / Drainage)	Autodesk Civil 3D	2024
Design Authoring (Non-BIM)	AutoCAD	2018 & Above
3D Coordination	Autodesk Navisworks	2024
4D Phase Planning	Bentley Synchro 4D	Latest
5D Cost Estimation	CostX	Latest
Visualization / AR	Unreal Engine / Twinmotion	Latest
Document Management	Aconex / ACC	Latest
Asset Management	FSI / ACC Build	Latest
Site Capturing	LiDAR 3D Scanning	Latest
GIS / Site Planning	ArcGIS Pro	Latest

Table 34 – Software Platforms

Hardware

Component	Specification
Processor (CPU)	13th Gen Intel Core i9 (32 CPUs), ~3.0 GHz or equivalent
GPU	NVIDIA GeForce RTX 4090 24 GB or equivalent
RAM	64 GB DDR5 (2x32 GB) 6000 MHz
Storage	1 TB NVMe SSD (primary) + 500 GB NVMe (secondary)
Monitor	Dual 27" monitors (2560x1440 or higher)
Cooler	Liquid cooling system
Network	High-speed internet connection (min. 100 Mbps)

Table 35 – Hardware Requirements

8 Deliverables Summary

8.1 Resources to be Prepared

Deliverable
Delivery team's Pre-appointment BIM Execution Plan
Delivery team's Post-appointment BIM Execution Plan (this document)
Master Information Delivery Plan (MIDP)
Model Production Delivery Table (MPDT)
Level of Information Need
Clash Matrix
QA/QC Check List
Federated Model
Coordination Report
Asset Register
Project Information Model
Asset Information Model
Contribute to Project Lessons Learnt

Table 36 – Deliverable Summary

8.2 Stakeholders Responsibilities

RACI Key: R = Responsible | A = Accountable | C = Consulted | I = Informed

#	Document Title	Client	SC/PMC	GC	[Main Contractor Name]	Sub-Contractors
01	Modon's Exchange Information Requirements	A/R	I	I	I	I
02	Pre-appointment BEP Template	A/R	I	I	I	I
03	Capability & Capacity Assessment Template	A/R	I	I	I	I
04	Post-appointment BEP Template	A/R	I	I	I	I
05	MIDP Template	A/R	I	I	I	I
06	MPDT Template	A/R	I	I	I	I

07	Information Management Assignment Matrix (RACI)	A/R	I	I	I	I
08	Information Uses	A/R	I	I	I	I
09	Level of Information Need	A/R	I	I	I	I
10	Clash Matrix	A/R	I	I	I	I
11	QA/QC Check List	A/R	I	I	I	I
12	CDE Folder Structure	A/R	I	I	I	I
13	Pre-appointment BIM Execution Plan	C	A	I	R	-
14	Post-appointment BIM Execution Plan (this BEP)	C	A	C	R	I
15	MIDP	C	A	C	R	I
16	MPDT	C	A	C	R	I
17	Level of Information Need	C	A	C	R	I
18	Clash Matrix	C	A	C	R	I
19	QA/QC Check List	C	A	C	R	I
20	Asset Register	C	A	C	R	I
21	BIM Progress Report	C	A	C	R	I
22	Coordination Report	C	A	C	R	I
23	Project Information Model	C	A	C	R	I
24	Federated Model	C	A	C	R	I
25	Asset Information Model	C	A	C	R	I

Table 37 – Deliverables Information Management Assignment Matrix (RACI)

9 Appendices

9.1 List of Appendices

Document Name	Document Code	Revision	Purpose
[Model Breakdown Structure]	[Appendix 01]	00	For Review
[QA/QC Check List Template]	[Appendix 02]	00	For Review
[Level of Information Need]	[Appendix 03]	00	For Review
[Title Block Template]	[Appendix 04]	00	For Review
[Model Production Delivery Plan Table (MPDT)]	[Appendix 05]	00	For Review
[Master Information Delivery Plan (MIDP)]	[Appendix 06]	00	For Review
[Master Clash Detection Matrix]	[Appendix 07]	00	For Review
[Datum File]	[Appendix 08]	00	For Review

Table 38 – List of Appendices

— End of Document —

This document has been prepared by [Main Contractor Name] as the appointed Main Contractor under [General Contractor Name]. All information contained herein is confidential and shall not be disclosed without prior written consent from [Client Name].

FOR USER

Appendix Package – Complete BIM Documentation Templates

The complete set of editable appendices referenced in this BIM Execution Plan is available as a **ready-to-use BIM documentation package**.

The package includes:

- Model Breakdown Structure
- QA/QC Checklist Template
- Level of Information Need (LOIN)
- Title Block Template
- Model Production Delivery Plan Table (MPDT)
- Master Information Delivery Plan (MIDP)
- Master Clash Detection Matrix
- Datum File
- Additional supporting BIM templates and documentation resources

Need the complete package?

Get the full **ISO 19650 BIM Execution Plan Appendix Package** with editable templates designed to support BIM project planning, information management, coordination, QA/QC and delivery requirements.

For package details, pricing and purchase:

Contact Email: [Appendix Package – Complete BIM Documentation Templates](#)

Website / Store: <https://vishnuverse60.gumroad.com/l/iso19650-bim-appendix-package>

This package is intended as a practical template/resource and should be reviewed and adapted to suit the specific project requirements, contractual obligations and information management standards applicable to each project.